

Low-density phase diagram of the two-dimensional Coulomb gas

J. M. Caillol and D. Levesque

Laboratoire de Physique Théorique et Hautes Energies, Université de Paris—Sud, 91405 Orsay, France

(Received 24 May 1985)

The phase diagram of the two-dimensional classical Coulomb gas is investigated by Monte Carlo simulations. We show that in the fluid phase this system undergoes the expected Kosterlitz-Thouless transition at very low densities and that the Kosterlitz-Thouless transition becomes a first-order phase transition between a dielectric gas and a conductive liquid at moderate densities. We give an estimate of the location of the critical point of this first-order transition.

I. INTRODUCTION

The two-dimensional Coulomb gas (CG) with logarithmic interactions has already benefited from numerous theoretical studies. Hauge and Hemmer¹ have considered the case of the system of point charges e and have shown that this system is unstable for temperatures lower than $T_i = e^2/2k_B$. This instability results from the collapse of the system by formation of neutral pairs of infinite negative energy.^{2,3} This collapse can be avoided by either adding to the logarithmic interaction a short-range repulsive potential, for instance, a hard-core interaction, or by placing the particles on the sites of a lattice. Kosterlitz and Thouless⁴ (KT) have studied these versions of the CG system and have established the existence of a continuous transition between a dielectric phase with finite dielectric constant ϵ and a conductive phase with an infinite ϵ . The predicted transition temperature is $T_c \leq e^2/4k_B$ at very low densities. The dielectric phase is composed of bounded particles forming dipoles. Below T_c all the particles are bounded in pairs and above T_i all the pairs are broken.

This transition presents an important theoretical interest: in two-dimensional systems such as helium films, XY magnets, etc., phase transitions without long-range order can occur by an unbinding of point defects similar to the unbinding of pairs in the CG system.^{5,6}

The CG model on a lattice and the XY model have already been studied by numerical simulations;^{7,8} these computations confirm many of the theoretical predictions, in particular, the value of T_c . In both studies the considered thermodynamical states correspond to low densities either of bounded pairs (CG model) or of vortices (XY model). For the lattice CG model only one value of the chemical potential has been considered. The aim of this article is to study the KT transition in the case of the fluid model of the CG gas in order to establish the variation of T_c with density over a wide domain.

The first section of this article is devoted to the results of numerical simulations in the very-low-density domain. Our results are similar to those of Saito *et al.*⁷ We show that the transition temperature is an increasing function of the density. The results of computations for moderate densities are reported in Sec. III; our main result is that the KT transition becomes a first-order transition between a dielectric gas and a conductive liquid. In the conclusion

(Sec. IV) we compare the characteristics of this first-order transition with the analogous one in the three-dimensional Coulomb systems.

II. PHASE DIAGRAM AT VERY LOW DENSITIES

The numerical simulation of Coulomb systems needs some caution because of the long-range character of the interaction. A common way of handling the long-range contribution of the interaction is to use periodic boundary conditions and to apply the Ewald-Kornfeld summation.⁹ We have chosen a different method, consisting of placing the particles on the surface of a sphere. This method is much less demanding on computer time than the Ewald method. It has already been used for numerical computations for the electron gas on a surface,¹⁰ the two-dimensional one-component plasma,¹¹ and the two-dimensional polar fluid.¹²

A. Description of the system

The system considered here is composed of $N^+ = N/2$ hard disks of diameter σ carrying a positive charge $+e$, and of $N^- (=N^+)$ hard disks of the same diameter carrying a negative charge $-e$. The particles are constrained to move on the surface S of a sphere of center 0 and radius R . The interaction potential between the particles i and j located at M_i and M_j on the sphere is

$$v_{ij} = -e_i e_j \ln(s_{ij}/L) + v_{ij}^{SR}, \quad (1)$$

where L is a length scale and s_{ij} is the length of the chord joining M_i and M_j :

$$s_{ij} = 2R \sin(\psi_{ij}/2), \quad (2)$$

with

$$\psi_{ij} = \arccos(\overline{OM_i} \cdot \overline{OM_j} / R^2), \quad 0 \leq \psi_{ij} \leq \pi. \quad (3)$$

v_{ij}^{SR} is a "hard-disk" potential, defined by

$$v_{ij}^{SR} = \begin{cases} \infty, & R\psi_{ij} \leq \sigma \\ 0, & R\psi_{ij} > \sigma. \end{cases} \quad (4)$$

The peculiar logarithmic interaction (1) has been shown to lead to the correct thermodynamic limit in the one-

component-plasma model.^{13,14}

The thermodynamical states of the system are defined by the two parameters $\Gamma = \beta e^2$ ($\beta = 1/k_B T$) and the packing fraction η equal to Ns/S , where s is the surface of a hard disk on the sphere [$s = 4\pi R^2 \sin^2(\sigma/4R)$]. For a large sphere $\sigma/R \ll 1$, we have $\eta \simeq \pi \rho \sigma^2/4$, where ρ is the number density.

From Eq. (1) we compute the potential energy:

$$V_N(\Omega_1, \Omega_2, \dots, \Omega_N) = Nu_0 + \sum_{\substack{i,j \\ i < j}} v_{ij}^{SR} - \sum_{\substack{i,j \\ i < j}} (e_i e_j / 2) \ln(1 - \cos \psi_{ij}), \quad (5)$$

with $u_0 = (e^2/4) \ln(2R^2/L^2)$ and Ω_i standing for (θ_i, ϕ_i) (θ_i, ϕ_i are spherical coordinates of M_i).

Most of the equilibrium properties of the system can be expressed in terms of the charge-charge correlation function $S(\Omega_1, \Omega_2)$,

$$S(\Omega_1, \Omega_2) = \left\langle \sum_{i,j} e_i e_j \delta(\Omega_1 - \Omega_i) \delta(\Omega_2 - \Omega_j) \right\rangle / R^4, \quad (6)$$

where $\delta(\Omega) = \delta(\cos \theta) \delta(\phi)$ and the angular brackets $\langle \dots \rangle$ refer to a canonical ensemble average. The charge neutrality of the system implies

$$\int S(\Omega_1, \Omega_2) d\Omega_2 = 0. \quad (7)$$

The excess internal energy per particle is then obtained from

$$\begin{aligned} U^{\text{ex}}/Ne^2 &= \frac{\langle V_N \rangle}{Ne^2} \\ &= \frac{u_0}{e^2} - \frac{1}{2Ne^2} \int d\Omega_2 R^4 S^d(\Omega_1, \Omega_2) \\ &\quad \times \ln(1 - \cos \psi_{12}), \end{aligned} \quad (8)$$

where $S^d(\Omega_1, \Omega_2)$ is defined by

$$S^d(\Omega_1, \Omega_2) = S(\Omega_1, \Omega_2) - \rho e^2 \delta(\Omega_1 - \Omega_2) / R^2. \quad (9)$$

The pressure P is calculated from the virial theorem:

$$\frac{\beta P}{\rho} = 1 - \Gamma/4 + \eta(g^{++}(\sigma) + g^{+-}(\sigma)), \quad (10)$$

where $g^{++}(\sigma)$ and $g^{+-}(\sigma)$ are the values of pair distribution functions of the like and unlike charges at contact.

The KT transition is the consequence of a change in the asymptotic decay of the correlations which changes from an exponential decay in the conductive phase to an algebraic decay in the dielectric phase.⁶ However, at very low densities the rather large statistical uncertainties which occur in the pair correlation functions at large distances did not allow us to distinguish unambiguously between these two types of decay. Furthermore, at the KT transition no observable singularities in the thermodynamical variables such as the specific heat are predicted. Finally, the quantity which seems to characterize without ambiguity the two phases is the dielectric constant ϵ , which is infinite in the conductive phase and finite in the dielectric phase.

For an infinite system, ϵ is given by the small- k limit of

the Fourier transform of the charge-charge correlation function $\tilde{S}(k)$,

$$1/\epsilon = \lim_{k \rightarrow 0} \left[1 - 2\pi\beta \frac{\tilde{S}(k)}{k^2} \right]. \quad (11)$$

If ϵ is infinite Eq. (11) implies that $\tilde{S}(k) \sim k^2/2\pi\beta$ for $k \rightarrow 0$. This low- k expression of $S(k)$ is equivalent to the so-called Stillinger-Lovett (SL) sum rule,^{15,16}

$$\int d^2r r^2 S(r) = -2/\pi\beta. \quad (12)$$

In the dielectric phase, where ϵ is finite, the SL sum rule is thus not satisfied. The breakdown of the SL sum rule can give several pieces of information on the nature of the decay of the correlations. In a recent recent publication Martin *et al.*¹⁶ deduced the SL sum rule from the BGY hierarchy under the main hypothesis that the two-body correlation functions fall off faster than $1/r^4$, and the three- and four-body correlation functions decay faster than $1/r^3$ as $r \rightarrow \infty$. *A fortiori*, the exponential decay of correlations in the conductive phase implies that ϵ is infinite. In the dielectric phase of the CG system it is likely that the two- three-, and four-body correlation functions behave at large distances as the corresponding correlation functions of a polar fluid of rigid pairs of oppositely charged particles. These correlation functions decay, respectively, as $1/r^4$, $1/r$, and $1/r^2$ (Refs. 17 and 18); these asymptotic behaviors are not compatible with the sufficient conditions postulated by Martin *et al.*, leading to a breakdown of the SL sum rule.

For a function defined on the surface of a sphere the Fourier expansion is a development on the complete set of spherical harmonics. For the charge-charge correlation function $S(\Omega_1, \Omega_2)$ [$\equiv S(\Omega)$], the coefficient S_n of this expansion is given by

$$S_n = \int d\Omega R^2 S(\Omega) P_n(\cos \theta), \quad (13)$$

where P_n is the n th Legendre polynomial. If $n(n+1) \ll N$ (cf. the Appendix), the SL sum rule is

$$S_n = n(n+1)/(2\pi\beta R^2). \quad (14)$$

For $n=1$ we have

$$S_1 = - \int d\Omega R^2 S(\Omega) (1 - \cos \theta) = \frac{1}{\pi\beta R^2}, \quad (15)$$

where the integration has been done with use of the neutrality conditions (7). S_1 is proportional to the mean-square fluctuations of the total dipolar moment $\mathbf{M}$:

$$\mathbf{M} = \sum_{i=1}^N e_i \overline{\mathbf{OM}_i}. \quad (16)$$

The relation between S_1 and $\langle \mathbf{M}^2 \rangle$ is

$$\langle \mathbf{M}^2 \rangle / R^2 = \langle 2 \sum_{i,j} e_i e_j \cos \phi_{ij} \rangle = 4\pi R^2 S_1. \quad (17)$$

Introducing the notation $\langle \mathbf{m}^2 \rangle = \Gamma \langle \mathbf{M}^2 \rangle / 4R^2 e^2$, we have the relation $\epsilon = 1/(1 - \langle \mathbf{m}^2 \rangle)$, which implies that $\langle \mathbf{m}^2 \rangle$ is equal to 1 in the conductive phase and that $\langle \mathbf{m}^2 \rangle$ is strictly smaller than 1 in the dielectric phase. In our computations we used $\langle \mathbf{m}^2 \rangle$ as the order parameter for characterizing the KT transition.

B. Results

In our Monte Carlo simulations we have used a system of 104 particles and we have computed the thermodynamic properties in the canonical ensemble for the three isochores: $\eta_1 = 5 \times 10^{-4}$, $\eta_2 = 5 \times 10^{-3}$, and $\eta_3 = 5 \times 10^{-2}$. For these densities the radii of the spheres are, respectively, 114.02σ , 36.05σ , and 11.4σ . Despite the relatively small number of particles, the surface of the system with the smallest density, η_1 , is comparable with the surface of the lattice system which was used by Saito *et al.*⁷

Figure 1 presents $\langle m^2 \rangle$ versus temperature Γ^{-1} for the three isochores η_1 , η_2 , and η_3 , with S ranging from 0.8 to 1.4. These values of $\langle m^2 \rangle$ have been obtained by Monte Carlo (MC) simulations involving about 10^6 configurations. For the three densities, Fig. 1 shows that $\langle m^2 \rangle$ has a sharp variation in a small domain of Γ where it goes from a value less than 1 to a constant value equal to 1. The domain of this variation is displaced toward the higher values of Γ as the density increases. In the dielectric phase, for a given Γ , $\langle m^2 \rangle$ is an increasing function of the density, which is typical of a polar fluid.

In the case of an infinitely large system and at the critical temperature T_c , ϵ is expected to jump from a finite value toward an infinite value¹⁹ and, consequently, $\langle m^2 \rangle$ has a finite jump at T_c .

For finite systems a roughening of the discontinuity of $\langle m^2 \rangle$ occurs. In order to estimate the finite-size effects, we have computed $\langle m^2 \rangle$ for systems of a different number of particles along the isochore η_2 . Obviously, the configuration integral of a system of two particles on a sphere of radius R at the packing fraction η is exactly computable:

$$\begin{aligned} Z^{(2)} &= \int d\Omega_1 R^2 \int d\Omega_2 R^2 e^{-\beta v_{12}} \\ &= 2\pi SL \Gamma \sigma^2 - \Gamma \frac{1 - \lambda^{\Gamma-2}}{\Gamma - 2}, \end{aligned} \quad (18)$$

where $\lambda = \sigma/2R$. For $\langle m^2 \rangle$ we obtain

$$\langle m^2 \rangle^{(2)} = 2\eta \Gamma \frac{\Gamma - 2}{1 - \lambda^{\Gamma-2}} \frac{1 - \lambda^{\Gamma-4}}{\Gamma - 4}. \quad (19)$$

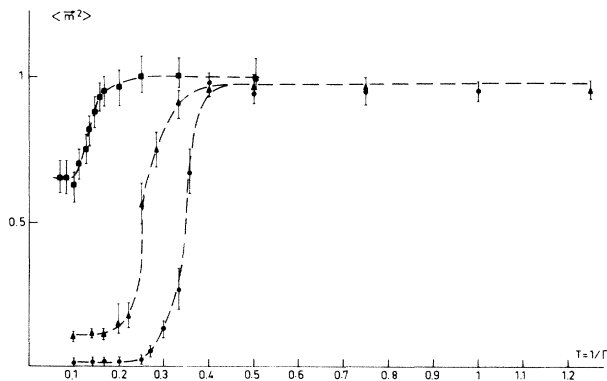


FIG. 1. $\langle m^2 \rangle$ versus the temperature for systems of 104 particles. Solid circles, isochore $\eta_1 = 5 \times 10^{-4}$; solid triangles, isochore $\eta_2 = 5 \times 10^{-3}$; solid squares, isochore $\eta_3 = 5 \times 10^{-2}$. The error bars are estimated variances of m^2 .

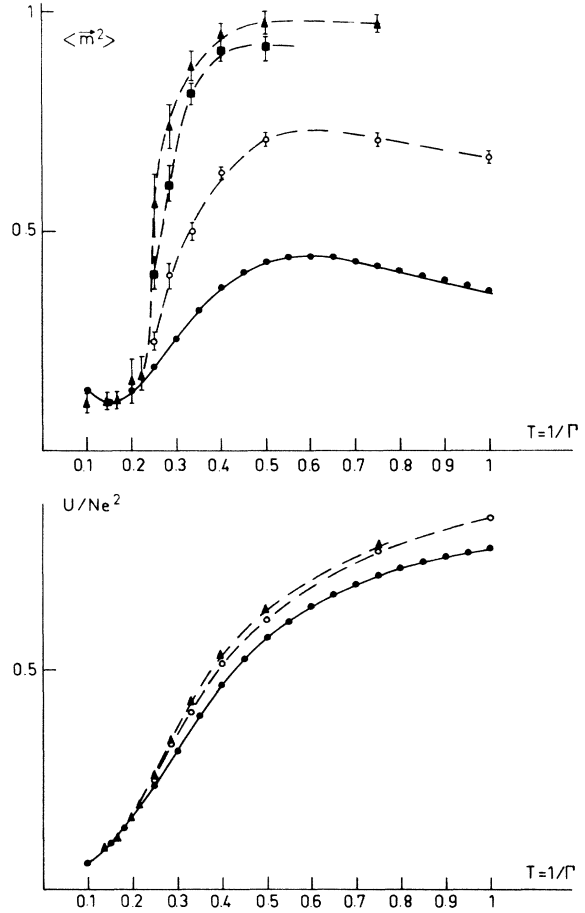


FIG. 2. (a) Size effects on $\langle m^2 \rangle$ vs Γ at the packing fraction $\eta_2 = 5 \times 10^{-3}$. Dots correspond to a system of $N=2$ particles; open circles, $N=8$; solid squares, $N=32$; solid triangles, $N=104$. (b) Size effects on the excess internal energy at the packing fraction $\eta_2 = 5 \times 10^{-3}$. The symbols have the same meaning as those of (a).

In Fig. 2 we plot $\langle m^2 \rangle$ versus the temperature Γ^{-1} along the isochore η_2 for $N=2, 8, 32$, and 104 . The main conclusion is that size effects are only important in the conductive phase.

Although the statistical uncertainties are small, the values of $\langle m^2 \rangle^{(N)}$ in the dielectric phase are indistinguishable for the four values of N . This result suggests that in the low-density dielectric phase the system behaves as an assembly of $N/2$ independent pairs.

In the conductive phase the SL sum rule is not fulfilled for systems of two and eight particles, but is almost satisfied as soon as N is larger than 32. This shows that for systems too small perfect screening cannot be obtained.

For values of $\Gamma \approx 4$ our results are compatible with a scaling of $1 - \langle m^2 \rangle^{(N)}$ given by $N^{-z(\Gamma)}$. Near the dielectric phase, for instance, $\Gamma=4$, the estimated value of z is ~ 0.2 . With such a value a system size of $\sim 10^5$ particles would be necessary to obtain $\langle m^2 \rangle \sim 0.9$.

Despite the roughening of the transition, a precise determination of the transition temperature is possible at low densities if we admit the results of the renormaliza-

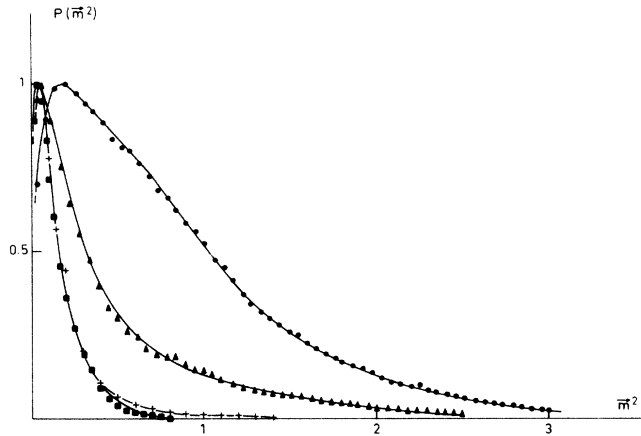


FIG. 3. Distribution of m^2 for the isochore $\eta_2 = 5 \times 10^{-3}$ and some values of Γ . Solid squares, $\Gamma=5$; crosses, $\Gamma=4.5$; solid triangles, $\Gamma=4$; solid circles, $\Gamma=3$.

tion analysis,^{4,6,19,20} which gives an expression of ϵ in the dielectric phase near the critical temperature:

$$\frac{\epsilon(\Gamma)}{\Gamma} = \frac{1}{4} - c \left[\frac{T_c - T}{T_c} \right]^{1/2} \quad (20)$$

($T < T_c$ and $c > 0$). The applicability of (20) for the determination of T_c is justified by the lack of size effects in the dielectric phase. For the two smallest densities the critical values of Γ_c are found to be 4 ± 0.1 for η_1 and 4.75 ± 0.25 for η_2 . Formula (20) is probably not applicable for the density $\eta_3 = 5 \times 10^{-2}$, but if it is used it gives $\Gamma_c \approx 10$.

We have found also a qualitative variation of the probability distribution of m^2 , $P(m^2)$, due to the KT transition. In Fig. 3 $P(m^2)$ is plotted for four values of Γ at the packing fraction η_2 . The very small values of m^2 are forbidden by the hard-core effects ($m^2 \gtrsim 2\eta\Gamma$ at low density). The distributions $P(m^2)$ in the conductive phase ($\Gamma=3$ and 4) present a tail going far beyond $m^2=1$, which is a consequence of the breaking of the pairs. This tail is absent in the dielectric phase ($\Gamma=5$). These results corroborate our previous estimate of the critical temperature.

Similar to the distribution $P(m^2)$, the evolution of m^2 as a function of the Monte Carlo samplings gives insight into the transition; this evolution is shown in Fig. 4. At $\Gamma=2.5$, m^2 fluctuates around 1 as expected in the conductive phase. At $\Gamma=4$, near the transition, we obtain metastability effects with an equal probability for the system to be in one of the two phases. At $\Gamma=5$, m^2 has the weak values typical of a dielectric phase, but, nevertheless, the system undergoes a brief fluctuation towards the metastable conductive phase.

The excess internal energy per particle obtained by MC simulations is reported in Table I for three isochores. For

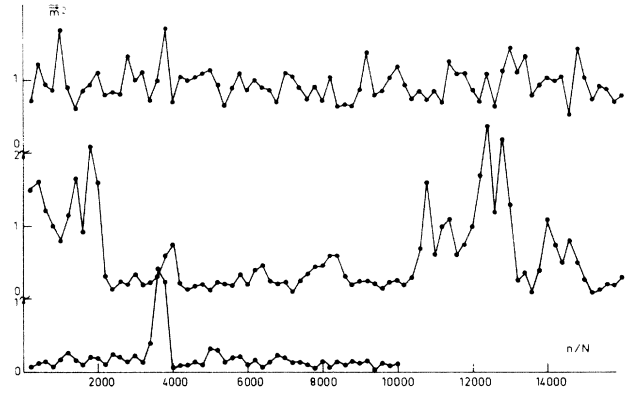


FIG. 4. Evolution of m^2 versus the number of configurations n for a system of $N=104$ particles at the packing fraction $\eta_2 = 5 \times 10^{-3}$. m^2 is defined as the partial average of m^2 over $100 \times N$ successive Monte Carlo steps. From top to bottom the curves correspond to values of Γ equal to 2.5, 4, and 5.

each phase a simple theory permits to reproduce these MC data.

At low temperatures, as we have already pointed out, the system may be described as an ensemble of $N/2$ independent pairs. In this approximation the partition function is

$$Q_N = \frac{1}{(N/2)!} \frac{1}{(\lambda^+ \lambda^-)^N} (Z^{(2)})^{N/2}, \quad (21)$$

where λ^+ and λ^- are the thermal de Broglie lengths of the particles and $Z^{(2)}$ is the configurational integral defined in (18). From (21) we obtain the excess internal energy,

$$\frac{U^{\text{ex}}}{Ne^2} = \frac{1}{2(\Gamma-2)} + \frac{1}{2} \frac{\lambda^{\Gamma-2} \ln \lambda}{1 - \lambda^{\Gamma-2}}, \quad (22)$$

where the unit of length has been taken equal to σ , and $L = \sigma$.

At high temperatures the system is a low-density plasma, the properties of which can be described by a straightforward extension to the two-dimensional case of the so-called DHLL + B2 theory.^{3,21} In this improved Debye-Hückel theory the hard-core effects are taken into account and the excess free energy F^{ex} is

$$\begin{aligned} \frac{\beta F^{\text{ex}}}{N} = & -\frac{\Gamma}{4} [\ln \Gamma + 2\gamma + 2 \ln(L/a) - 1] \\ & - \frac{\rho}{2} (B_{++} + B_{+-}), \end{aligned} \quad (23)$$

where $\gamma = 0.577 \dots$ is the Euler constant and a is the average distance between identical particles, $a = (2/\pi\rho\sigma^2)^{1/2}$. B^{++} and B^{+-} are given by³

$$B^{+(\pm)} = \frac{1}{2} \int d^2r \{ e^{-\beta[v_{\text{SR}}(r) + v_{\text{DH}}^{+(\pm)}(r)]} - 1 + \beta v_{\text{DH}}^{+(\pm)}(r) - \frac{1}{2} [\beta v_{\text{DH}}^{+(\pm)}(r)]^2 \}, \quad (24)$$

where $v_{\text{DH}}^{+(\pm)}(r)$ is the screened Debye-Hückel potential for the two-dimensional plasmas, namely

$$\beta v_{\text{DH}}^{+(\pm)}(r) = \pm \Gamma K_0(r/\lambda_D), \quad \lambda_D^{-1} = \sqrt{2\pi\rho\Gamma}, \quad (25)$$

TABLE I. U_1 , U_2^{ex} , and U_3^{ex} are, respectively, the excess internal energies of a system of $N=104$ particles along the three isochores $\eta_1=5\times 10^{-4}$, $\eta_2=5\times 10^{-3}$, and $\eta_3=5\times 10^{-2}$. The zero of energy is determined by the choice $L=\sigma$; n_1 , n_2 , and n_3 are the number of MC configurations which have been generated for the evaluation of V^{ex} .

Γ	U_1^{ex}/Ne^2	U_2^{ex}/Ne^2	U_3^{ex}/Ne^2	n_1 (10^{-6})	n_2 (10^{-6})	n_3 (10^{-6})
0.8	1.46(1)	0.932(2)		0.6	0.8	
1.0			0.430(1)			0.6
1.333	1.260(5)	0.786(2)		0.6	0.4	
2.0	0.972(5)	0.639(2)	0.303(2)	1.4	0.6	0.6
2.5	0.726(3)	0.534(2)		1.6	1.6	
2.777	0.597(9)			1.4		
3	0.48(1)	0.432(2)	0.228(2)	2.0	1.6	0.8
3.333	0.375(9)				1.6	
3.5		0.343(3)			1.6	
3.7037	0.292(5)			0.8		
4	0.262(5)	0.258(3)	0.167(1)	1.2	2.0	0.8
4.5		0.195(2)			1.2	
5	0.172(4)	0.167(2)	0.126(1)	0.8	1.0	0.8
6	0.125(2)	0.114(2)	0.086(1)	0.8	1.0	0.8
6.5			0.069(2)			1.2
7	0.099(1)	0.093(1)	0.057(2)	0.8	1.0	1.2
7.5			0.039(2)			1.2
8			0.031(2)			1.2
8.5			0.021(2)			1.2
9			0.006(2)			2.0
10	0.062(1)	0.041(2)	-0.007(1)	1.0	1.0	1.2
12			-0.038(1)			1.4
14			-0.063(1)			1.4

and using the fact that $\int_0^\infty x K_0^2(x) dx = \frac{1}{2}$, we obtain

$$B_{++} + B_{+-} = -\pi\sigma^2 - \frac{\pi\Gamma^2\lambda_D^2}{2} + 2\pi \int_\sigma^\infty r \{ \cosh[rK_0(r/\lambda_D)] - 1 \} dr.$$

From (23) the excess internal energy is calculated as

$$U^{\text{ex}}/Ne^2 = -\frac{1}{4} [\ln\Gamma + 2\gamma + 2\ln(L/a)] + \frac{1}{8} - \pi\rho \int_\sigma^\infty dr r \sinh \left[\Gamma K_0 \left(\frac{r}{\lambda_D} \right) \right] \left[K_0 \left(\frac{r}{\lambda_D} \right) - \frac{\sqrt{\Gamma}}{a} r K_1 \left(\frac{r}{\lambda_D} \right) \right]. \quad (26)$$

Figure 5 illustrates the fairly good agreement between the theory based on independent pairs and the MC results for temperatures corresponding to the dielectric phase. For temperatures slightly above the critical temperature ($3 \leq \Gamma \leq \Gamma_c$), the agreement is still good due to the weak concentrations of free particles. The internal energies computed for a finite system and for a system in the thermodynamic limit [$N \rightarrow \infty$ and $\lambda=0$ in (22)] are identical in these two domains of temperature. Below $\Gamma=3$ the theory fails because the thermodynamic limit of the internal energy diverges at $\Gamma=2$.

Satisfactory agreement between the DHLL + B2 theory and the MC results is found for values of $\Gamma < 2$. For values of $\Gamma > 2$ the discrepancy between theory and MC results is a consequence of the divergence occurring in the DHLL + B2 theory at $\Gamma=4$.

We have attempted to describe the conductive phase as a two-component mixture of pairs and free charged particles following Hansen and Viot,³ the concentration of pairs and free particles being obtained from a minimization of the free energy. If the free energies for the components of the mixture are those given by the two previous theories, this description fails due to the divergences occurring in both free energies.

As already noted by Saito and Müller-Krumbhaar⁶ in their study of the lattice CG, the finite-size effects on the internal energy are very small in the dielectric and conducting phases (see Fig. 2). For instance, the energies per particle for systems of eight and 104 particles differ by less than 2% in the range of temperatures that we have studied. The theoretically predicted singularity in the specific heat⁶ is very weak and certainly unobservable in a

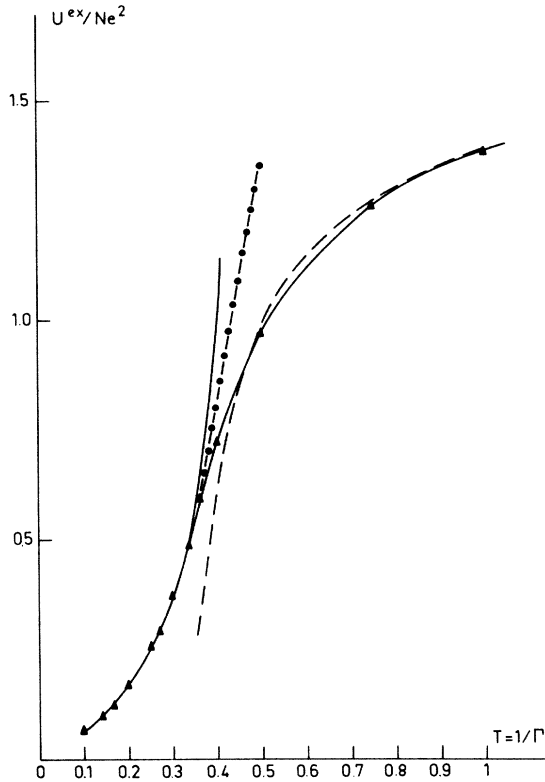


FIG. 5. Excess internal energy versus temperature for the isochore $\eta_1 = 5 \times 10^{-4}$. Solid triangles correspond to the MC results (104 particles) and the dashed line to the DHLL + B2 theory. The dotted and solid lines refer to the model of independent pairs in the cases of a finite and infinite system, respectively.

MC simulation. However, the entropy production due to the breaking of the pairs in the conducting phase induces a maximum in the specific heat at values of Γ below Γ_c . Our results are summarized in Fig. 6, and they are similar to those obtained in Ref. 7 for the lattice CG.

III. FIRST-ORDER TRANSITION

In this section we present the results of numerical simulation for the CG fluid in the domain of packing fractions $0.05 \leq \eta \leq 0.2$ and the range of values of the parameter Γ , $6 \leq \Gamma \leq 15$. The numerical simulations show that in this region of the phase diagram there is a first-order gas-liquid transition whose critical point is located at $\eta_c \simeq 0.11$, $\Gamma_c \simeq 11.5$, and $P_c/e^2 \simeq 1.8 \times 10^{-3}$ (see Fig. 7). The transition takes place between a dielectric gas and a conductive liquid, which is consistent with the transformation of the continuous KT transition in this first-order transition.

In the canonical ensemble for finite values of N , a gas-liquid transition generates van der Waals loops for the isotherms below the critical temperature.²² In the CG fluid these van der Waals loops happen to occur at very low pressures ($P/e^2 \equiv P^* \sim 10^{-3}$). This fact makes a precise estimation of P^* along these loops through use of the virial theorem difficult. In formula (10), if P^* is very small (10^{-3}), there is an almost exact cancellation between the two contributions $1 - \Gamma/4$ and $\eta[g^{++}(\sigma) + g^{+-}(\sigma)]$, and thus a determination of $g^{++}(\sigma)$ and $g^{+-}(\sigma)$ with an error of 1% is required in order to calculate P^* with a precision of 10%. Because very long calculations are needed to obtain a precision of 1% on the values of the two-body correlation functions, it has appeared useful to study the gas-liquid transition of the CG fluid by simulations in the isobaric ensemble (NPT ensemble). The advantages of the use of this ensemble are that the pressure has a given value, and that for temperatures below the critical temperature it is easy to locate the gas-liquid transition by the occurrence of metastable states in the two phases.

The MC procedure for the numerical simulations in the NPT ensemble has been described with many details by Wood²³ for the system of particles interacting via a hard-core potential. Here we summarize briefly this procedure.

With the notation used in Sec. I, the isobaric partition function Δ is written as

$$\begin{aligned} \Delta &= \frac{1}{N+1N^{-(\lambda+\lambda^-)N/2}} c \int_0^\infty e^{-\beta PS} dS \int R^{2N} \prod_{i=1}^N d\Omega_i \\ &\quad \times \exp \left[\beta \sum_{\substack{i,j \\ i < j}} \frac{e_i e_j}{2} \ln \left[\frac{2R^2}{L^2} (1 - \cos \psi_{ij}) \right] \right] \exp \left[-\beta \sum_{\substack{i,j \\ i < j}} v^{SR} (R \psi_{ij}) \right] \\ &= \frac{1}{N+1N^{-(\lambda+\lambda^-)N/2}} c \int_0^\infty e^{-\beta PS} Z(N, \beta, S) dS \\ &= \frac{(L^2/2)^{\Gamma N/4}}{N+1N^{-(\lambda+\lambda^-)N/2}} c \\ &\quad \times \int_0^\infty e^{-\beta PS} \left[\frac{S}{4\pi} \right]^{N(1-\Gamma/4)} \int \prod_{i=1}^N d\Omega_i \exp \left[\beta \sum_{\substack{i,j \\ i < j}} \frac{e_i e_j}{2} \ln(1 - \cos \psi_{ij}) \right] \exp \left[-\beta \sum_{\substack{i,j \\ i < j}} v^{SR} \left[\left[\frac{S}{4\pi} \right]^{1/2} \psi_{ij} \right] \right]. \end{aligned}$$

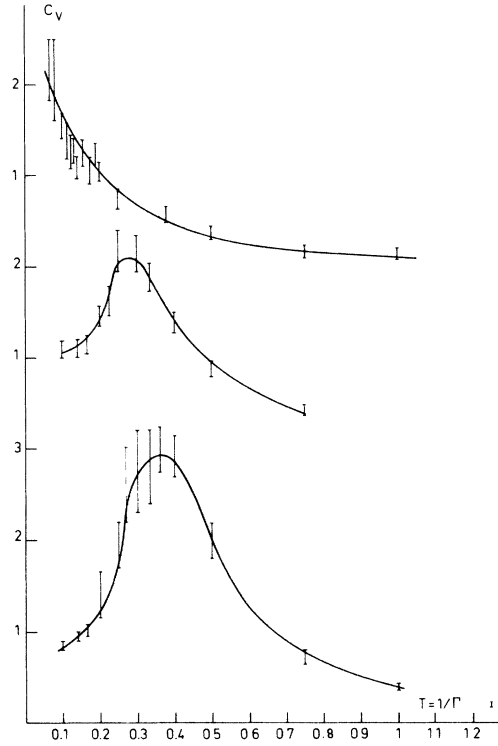


FIG. 6. Specific heat versus temperature for the three isochores $\eta_1 = 5 \times 10^{-4}$, $\eta_2 = 5 \times 10^{-3}$, and $\eta_3 = 5 \times 10^{-2}$ (from bottom to top). Solid lines are interpolations through the simulations results.

c is a constant having the dimension of S^{-1} , which makes Δ dimensionless (another possible choice for c is βP , as proposed by Sack²⁴).

At pressure P and a fixed value of Γ , the average value of S is calculated by

$$\langle S \rangle = \frac{\int_0^\infty dS S e^{-\beta P S} Z(N, \beta, S)}{\int_0^\infty dS e^{-\beta P S} Z(N, \beta, S)}.$$

The evaluation of $\langle S \rangle$ is made by the Metropolis algorithm.²⁵ In this algorithm the elementary step consists of sampling one of the variables Ω_i or the surface area S according to the probability $c_0 e^{-\beta P S} e^{-\beta V_N}$ (c_0 is a normalization constant). The frequency of the sampling of S with respect to that of the variables Ω_i has been chosen empirically. After several trials it has appeared that a relative frequency of ~ 10 – 100 between the samplings of the variables Ω_i and of the surface area S permits a fast evolution of S toward the domain of its probable values. For a relative frequency of $\simeq 1$, the sampling of S would be inefficient in the case where strongly bound pairs are present in the system. Indeed, samplings of S which result in a decrease of the value of S will not be accepted due to the overlap of the particles forming pairs. The average $\langle S \rangle$ is calculated with an uncertainty of 10% if 10^5 samplings of S are performed. For a value P_0 of the

pressure, the average value of S computed in the NPT ensemble is equal to the value of S giving a pressure P_0 in the canonical ensemble only in the thermodynamic limit. Wood²³ has shown that, for a finite value of N , the pressure computed from the canonical ensemble has the value P_0 only if the surface S in the canonical ensemble has a value $\bar{S}$ corresponding to the most probable value of S in the NPT ensemble where P is equal to P_0 . Then for an identical value of the pressure in the two ensembles, the difference between the values of S in the two ensembles will be the difference between $\langle S \rangle$ and $\bar{S}$. This difference is very difficult to estimate, but it is reasonable to assume that it will be appreciable if the variance of S , i.e., the compressibility, is large.

The results of our calculations for the isotherms $\Gamma = 6, 10, 11, 12$, and 15 are given in Table II and Fig. 7. For the isotherms $\Gamma = 6$ and 12 we have results of simulations in the NPT and canonical ensembles; the other isotherms are computed using the NPT ensemble. In the canonical-ensemble simulations, the isotherm $\Gamma = 6$ has no van der Waals loop, but a loop exists at $\Gamma = 12$. For $\Gamma = 6$ there is rather good agreement between the pressures and the packing fractions obtained from the simulations in the two ensembles, except for the low pressures, where the packing fractions differ by 20% or 30%. In the case of $\Gamma = 12$, we expect large differences between the two sets of P and η resulting from the simulations in the two ensembles due to the fact that the two ensembles are not equivalent in the two-phase region, even in the thermodynamic limit. However, the differences look very large, indicating a possible systematic error in the difficult evaluation of the contact values $g_{++}(\sigma)$ and $g_{+-}(\sigma)$ of the two-body correlation functions.

In the NPT ensemble the calculation of $\langle \eta \rangle$ reveals the existence of metastable states for $\Gamma > 11$. At $\Gamma = 12$ the

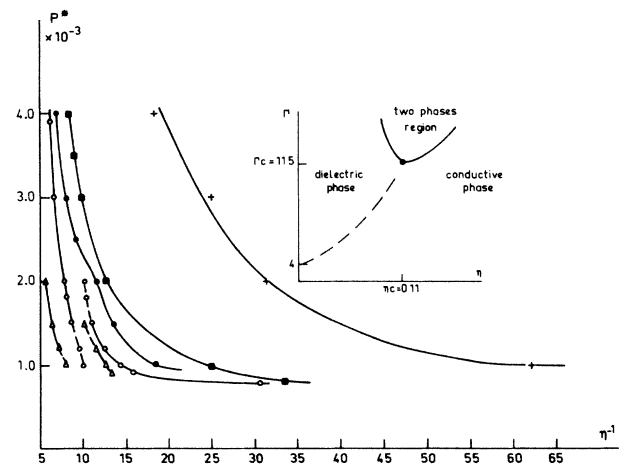


FIG. 7. Equations of state of the CG fluid in the vicinity of the first-order transition. Crosses, isotherm $\Gamma = 6$; squares, $\Gamma = 10$; dots, $\Gamma = 11$; circles, $\Gamma = 12$; triangles, $\Gamma = 15$. The dashed parts of the isotherms correspond to the metastable states. In the inset we have sketched the phase diagram over the entire range of densities and temperatures that we have explored.

TABLE II. Equation of state in the vicinity of the first-order transition. First column, pressure P^* ; second column, packing fraction $\langle \eta \rangle$ [for $\Gamma=12$ and 15 —left part, estimated value for the gas phase; right part, estimated value for the liquid phase (the parentheses indicate the phase which seems unstable)]; third column, $\langle m^2 \rangle$; fourth column, Γ ; fifth column, n total number of samplings on the variables Ω_i ; sixth column, n_s , the total number of samplings on the variable S ; the last column gives the type of the statistical ensemble used in the MC simulations. In the canonical-ensemble calculations, typically 10^6 MC configurations were generated. Typical uncertainties on $\langle \eta \rangle$ calculated by the NPT ensemble are ± 0.005 — ± 0.01 and $\pm 0,1$ on $\langle m^2 \rangle$.

P^* (10^{-3})	$\langle \eta \rangle$	$\langle m^2 \rangle$	Γ	n (10^{-6})	n_s (10^{-5})	Ensemble
6.60 $\pm$ 0.05	0.08	1.0	6	4.0	0.4	NVT
4.40 $\pm$ 0.05	0.06	1.0				NVT
3.95	0.054					NPT
3.92 $\pm$ 0.03	0.05	1.0				NVT
3.15 $\pm$ 0.03	0.004	0.8				NVT
3.0	0.04	0.8		2.0	0.8	NPT
2.45 $\pm$ 0.05	0.03	0.8				NVT
2.00	0.033	0.5				NPT
1.62 $\pm$ 0.04	0.02	0.4		3.6	1.4	NVP
1.00	0.016	0.4				NPT
0.95 $\pm$ 0.03	0.01	0.3				NVT
4.0	0.13	1.0	10	2.2	0.9	NPT
3.5	0.11	0.9		2.0	0.8	NPT
3.0	0.10	0.9		2.4	1.0	NPT
2.0	0.08	0.8		2.0	0.8	NPT
1.0	0.04	0.6		2.4	1.0	NPT
0.8	0.03	0.5		2.4	2.0	NPT
4.0	0.15	1.0	11	1.0	0.5	NPT
3.0	0.13	1.0		1.8	0.7	NPT
2.5	0.11	1.0		1.8	1.4	NPT
2.0	0.085	0.9		3	2.4	NPT
1.5	0.07	0.8		3.0	2.4	NPT
1.0	0.055	0.8		1.8	1.5	NPT
3.9	0.16		12	1.6	0.2	NPT
3.0	0.15			4.0	1.6	NPT
2.3	0.13					NVT
2.0	(0.095) 0.125	(1.0) 1.0		(1.2) 2.0	(1.0) 2.0	NPT
1.8	(0.095) 0.125	(1.0) 1.0		(1.5) 1.5	(1.2) 1.2	NPT
1.5	0.09	0.9 1.0		5.0 5.0	50.0 50.0	NPT
1.2	0.08	(0.10) 0.9 (1.0)		5.0 (4.0)	0.5 (0.4)	NPT
1.0	0.07	(0.10) 0.9 (1.0)		6.0 (4.0)	0.6 (0.4)	NPT
0.97 $\pm$ 0.06	0.115					NVT
0.9	0.065 (0.11)	0.9 (1.0)		5.0 (2.5)	0.5 (0.25)	NPT
0.8	0.033			2.0	0.2 (0.25)	NPT
0.75 $\pm$ 0.04	0.08	0.8				NVT
0.74 $\pm$ 0.07	0.04	0.6				NVT
0.72 $\pm$ 0.04	0.04	0.9				NVT
0.72 $\pm$ 0.04	0.06	0.8				NVT
0.66 $\pm$ 0.07	0.1	0.9				NVT
0.39 $\pm$ 0.04	0.01	0.2				NVT
2.0	0.19	1.0	15	7.6	6.0	NPT
1.5	(0.10) 0.15	(1.0) 1.0		(3.9) 6.4	(1.6) 2.5	NPT
1.2	0.085	0.9 1.0		4.5 8.0	1.8 3.2	NPT
1.0	0.08	(0.13) 0.7 (1.0)		9.2 (7.8)	3.7 (3.1)	NPT
0.9	0.075	(0.12) 0.7 (1.0)		4.0 (2.4)	1.6 (1.0)	NPT

metastability region corresponds to $0.9 \times 10^{-3} < P^* < 2.0 \times 10^{-3}$. The metastability is clearly seen in Fig. 8, giving the evolution of the value of η in the course of the Monte Carlo sampling for the pressure $P^* = 1.2 \times 10^{-3}$. In this evolution there are two regions of stability corre-

sponding to the values of $\eta \sim 0.10$ and ~ 0.08 . These values of η are interpreted as those of the packing fractions of the liquid and gas phases. For this thermodynamic state the largest number of generated MC configurations corresponds to the low-density configurations,



FIG. 8. Evolution of η versus the number of samplings of the surface area S at the pressure $P^* = 0.8 \times 10^{-3}$ (bottom) and $P^* = 1.2 \times 10^{-3}$ (top) for $\Gamma = 12$. $\bar{\eta}$ is a partial average of η over 500 successive samplings of the area S .

which indicates a probable stability of the gas phase.

A similar analysis of the MC results can be done for the range of pressures $0.9 \times 10^{-3} < P^* < 2.0 \times 10^{-3}$. For $P^* > 1.5 \times 10^{-3}$ the most probable values of η correspond to these of the liquid phase (see Table II).

Again at $\Gamma = 12$ the evolution of η is given for $P^* = 0.8 \times 10^{-3}$ in Fig. 8. For this pressure, the evolution of η presents a monotonic decrease which is followed by fluctuations around the value $\eta \sim 0.03$. Thus this thermodynamic state is clearly in the gas phase. For the isotherm $\Gamma = 15$ the metastabilities are more pronounced; for instance, at $P^* = 1.2 \times 10^{-3}$ and 1×10^{-3} , it was impossible to obtain a transition between the low- and high-density states in spite of the $\sim 3.0 \times 10^5$ samplings of the surface S . For $P^* = 1.5 \times 10^{-3}$ the system goes from its low-density configurations to its high-density configurations after 1.6×10^5 samplings of S . The inverse evolution is not observed even after 2.5×10^5 samplings. On the contrary, at $P^* = 0.9 \times 10^{-3}$ the transition between the high densities and the low densities takes place after 10^5 samplings of S .

The isotherm $\Gamma = 15$ corresponds to a temperature below the critical temperature, and the pressure of the equilibrium between the two phases is about $P^* \sim 1.2 \times 10^{-3}$. At this pressure the gas phase has the packing fraction $\eta_g \sim 0.085 \pm 0.01$ and the liquid phase has a packing fraction $\eta_l \sim 0.14 \pm 0.01$. The isotherm $\Gamma = 12$ is near the critical temperature, and the equilibrium between the two phases corresponds to $P^* \sim 1.5 \times 10^{-3}$ with $\eta_g \sim 0.09 \pm 0.01$ and $\eta_l \sim 0.115 \pm 0.01$. For $\Gamma = 11$ and 10 we have observed no metastable states. From these NPT ensemble results we estimate the critical point to be $\eta_c \sim 0.11$, $\Gamma_c \sim 11.5$, and $P_c^* \sim 1.8 \times 10^{-3}$. From the computation of $\langle m^2 \rangle$ we conclude in favor of the dielectric nature of the gas phase in the low-density region. For the isotherms $\Gamma = 6, 10$, and 11 from the re-

sults reported in Table II, we can estimate the density of the KT transition. The transitions take place for $\eta \sim 0.04$ at $\Gamma = 6$, $\eta \sim 0.06$ at $\Gamma = 10$, and $\eta \sim 0.07$ at $\Gamma = 11$. The evaluation of the packing fraction of the KT transition at $\Gamma = 10$ confirms our previous estimate of the temperature of the transition done on the isochore $\eta_2 = 0.05$ (see Sec. I).

For $\Gamma = 12$ and 15 at the pressures corresponding to the equilibrium between the two phases and at the packing fractions η_g , $\langle m^2 \rangle$ has values around 0.9 and the statistical uncertainties do not permit the unambiguous identification of the gas phase as being a dielectric fluid. However, for the low pressures at $\Gamma = 12$ and 15 there is no doubt concerning the dielectric nature of the gas phase. For all pressures in the liquid region the values of $\langle m^2 \rangle$ correspond to a conductive fluid.

IV. CONCLUSION

The main results that we have obtained in our numerical simulations of the two-dimensional Coulomb fluid are a rather precise determination of the critical temperatures of the KT transition at very low densities and the locations of the gas-liquid transition. This gas-liquid transition, which is an extension of the KT transition in the moderate-density region, is due to the presence of the short-range interaction v^{SR} , which is needed in order to avoid the collapse of the system.

Our conclusions concerning the phase diagram are summarized in the inset of Fig. 7, where the domain of the different phases is drawn in the plane of the variables Γ and η ; our MC simulations suggest that the KT critical line and the gas-liquid coexistence line join at the critical point, but no firm conclusion can be drawn on this result due to numerical uncertainties.

The phase diagram of the two-dimensional CG fluid

presents many analogies with the phase diagram²⁶ predicted for the three-dimensional systems of charged hard spheres. The more recent estimation of the critical point of the liquid-gas transition of this system is very similar to that obtained in our work, but obviously in the classical three-dimensional Coulomb system the gas is a conductive phase because the Coulomb interaction e^2/r is not able to create stable pairs.

ACKNOWLEDGMENTS

We are very indebted to Professor J. Lebowitz, Professor J. P. Hansen, and Professor J. J. Weis for their continuous interest in this work during its realization. We have also benefited from a very important suggestion of Professor B. Jancovici and from a fruitful discussion with Professor J. M. Kosterlitz. Parts of the calculations have been made possible through an allocation of time on the Cray-1S computer of the Centre de Calcul Vectoriel pour la Recherche at Palaiseau (France). According to a disposition concerning this allocation, we are required to state

$$\delta V_n^{\text{ext}}(\Omega_0) = - \int d\Omega' R^2 \frac{\delta a}{2} P_n(\cos\theta') \ln[2R^2(1 - \cos\{\arccos[\overline{OM}_0 \cdot \overline{OM}' / (|\overline{OM}_0| |\overline{OM}'|)]\})], \quad (\text{A1})$$

where we have made the simplifying choice $L = \sigma$. We calculate δV_n^{ext} by expanding the logarithm in (A1) on the complete set of the Legendre polynomials and by using the addition theorem and the orthogonality properties of these polynomials,

$$\delta V_n^{\text{ext}}(\Omega_0) = 2\pi R^2 \delta a \frac{P_n(\cos\theta_0)}{n(n+1)}. \quad (\text{A2})$$

The external potential induces a nonuniform charge density $\delta\rho_n^i$, which, because $\beta\delta a \ll 1$, can be computed by linear-response theory,

$$\delta\rho_n^i(\Omega_0) = -\beta \int d\Omega' R^2 S(\Omega_0, \Omega') \delta V_n^{\text{ext}}(\Omega'). \quad (\text{A3})$$

In (A3) $S(\Omega_0, \Omega')$ is the charge-charge correlation function of the plasma in the absence of the external potential and depends only on the relative angle

$$\Omega = \arccos[\overline{OM}_0 \cdot \overline{OM}' / (|\overline{OM}_0| |\overline{OM}'|)];$$

it may be expanded on the $P_n(\cos\Omega)$ polynomials, and after a short calculation we find

$$\delta\rho_n^i(\Omega_0) = -2\pi R^2 \delta a \frac{S_n}{n(n+1)} P_n(\cos\theta_0), \quad (\text{A4})$$

with

$$S_n = \int d\Omega R^2 P_n(\cos\theta) S(\Omega). \quad (\text{A5})$$

The external charge density $\delta\rho_n^{\text{ext}}$ has a constant sign in zones of characteristic length $l = \pi R/n$. If l is much larger than the average distance between particles, $a_0 = 1/\sqrt{\pi\rho}$, the plasma will rearrange in such a way as to screen perfectly the external charge density,

$$\delta\rho_n^i + \delta\rho_n^e = 0, \quad (\text{A6})$$

that "Les moyens de calcul utilisés ont été attribués par le Conseil Scientifique du Centre de Calcul Vectoriel pour la Recherche." Laboratoire de Physique Théorique et Hautes Energies is a Laboratoire associé au Centre National de la Recherche Scientifique (France).

APPENDIX

In this appendix we give a short derivation of the Stillinger-Lovett sum rule for a two-dimensional plasma ($\Gamma < \Gamma_c$) lying on the surface of a sphere of center 0 and radius R and whose particles interact via the logarithmic potential defined in Eq. (1).

The macroscopic derivation given by Stillinger-Lovett¹⁵ in the case of a planar system can be transposed to our spherical geometry as follows: let $\delta\rho_n^{\text{ext}} = \delta a P_n(\cos\theta)$ be an external charge distribution on the sphere, where δa is a small constant ($\beta\delta a \ll 1$) and $P_n(\cos\theta)$ is the n th Legendre polynomial. The charge density $\delta\rho_n^{\text{ext}}$ generates an external electrostatic potential δV_n^{ext} ,

which leads to

$$S_n = \frac{n(n+1)}{2\pi\beta R^2}. \quad (\text{A7})$$

It is worth noting that the condition $l \gg a_0$ is equivalent to $n(n+1) \ll N$ and, thus, the perfect screening is possible only for systems with a large enough number of particles. Since in the plasma phase $S(\Omega)$ decays exponentially with a correlation length $\xi = R\Omega_\xi$, for a sufficiently large sphere, Ω_ξ will be small and we can estimate (A5) from a Taylor expansion of $P_n(\cos\theta)$ around $\theta = 0$:

$$P_n(\cos\theta) \approx 1 + n(n+1)(\cos\theta)/2.$$

We obtain

$$S_n \sim \frac{n(n+1)}{2} \int d\Omega R^2 S(\Omega) (\cos\theta - 1), \quad (\text{A8})$$

where we have used the neutrality equation (7).

The low- k behavior of the charge-charge structure factor in a plane is

$$\tilde{S}(k) \sim -\frac{k^2}{4} \int d^2r r^2 S(r). \quad (\text{A9})$$

For a large sphere we can relate the spherical and the planar geometries by identifying $R^2 d\Omega \simeq d^2r$ and $2R^2(1 - \cos\theta) = r^2$. From these two relations for low k and small n , we have

$$\tilde{S}(k) = S_n, \quad k^2 = n(n+1)/R^2. \quad (\text{A10})$$

Inserting (A10) into (A7), we obtain $\tilde{S}(k) \sim k^2/2\pi\beta$, which is the usual form of the SL sum rule in the plane (see Sec. I).

- ¹E. M. Hauge and P. C. Hemmer, *Phys. Norv.* **5**, 209 (1971).
²C. Deutsch and M. Lavaud, *Phys. Rev. A* **9**, 2598 (1974).
³J. P. Hansen and P. Viot, *J. Stat. Phys.* **38**, 823 (1985); P. Viot, Thèse de Troisième cycle, Université Paris VI (unpublished).
⁴J. M. Kosterlitz and D. J. Thouless, *J. Phys. C* **6**, 1181 (1973); J. M. Kosterlitz, *ibid.* **7**, 1046 (1974); **10**, 3753 (1977).
⁵J. V. José, L. P. Kadanoff, S. Kirkpatrick, and D. R. Nelson, *Phys. Rev.* **16**, 1217 (1977).
⁶D. R. Nelson, in *Phase Transitions and Critical Phenomena*, edited by C. Domb and M. S. Green (Academic, New York, 1976), Vol. 7.
⁷Y. Saito and J. Müller-Krumbhaar, *Phys. Rev. B* **23**, 308 (1980).
⁸J. Tobochnik and G. V. Chester, *Phys. Rev. B* **20**, 3761 (1979).
⁹S. W. de Leeuw, J. W. Perram, and E. R. Smith, *Proc. R. Soc. London, Ser. A* **373**, 27 (1980).
¹⁰J. P. Hansen, D. Levesque, and J. J. Weis, *Phys. Rev. Lett.* **43**, 979 (1979).
¹¹J. M. Caillol, D. Levesque, J. J. Weis, and J. P. Hansen, *J. Stat. Phys.* **28**, 325 (1982).
¹²J. M. Caillol, D. Levesque, and J. J. Weis, *Mol. Phys.* **44**, 733 (1981).
¹³J. M. Caillol, *J. Phys. (Paris) Lett.* **42**, L245 (1981).
¹⁴K. Eburu and K. Iguchi (unpublished).
¹⁵F. Stillinger and R. Lovett, *J. Chem. Phys.* **48**, 3858 (1968); **49**, 1991 (1968).
¹⁶P. A. Martin and C. Gruber, *J. Stat. Phys.* **31**, 691 (1983).
¹⁷J. S. Hoyer and G. Stell, *J. Chem. Phys.* **65**, 18 (1976).
¹⁸D. Chandler, *J. Chem. Phys.* **67**, 1113 (1977).
¹⁹T. Ohta and K. Kawasaki, *Prog. Theor. Phys.* **60**, 365 (1978).
²⁰A. P. Young, *J. Phys. C* **11**, L453 (1978).
²¹H. Falkenhagen and W. Ebeling, in *Ionic Interactions*, edited by S. Petrucci (Academic, New York, 1971), Vol. 1, and references cited therein.
²²T. L. Hill, in *Statistical Mechanics* (McGraw-Hill, New York, 1956), Chap. 7.
²³W. W. Wood, in *Physics of Simple Liquid*, edited by H. W. V. Temperley, J. S. Rowlinson, and G. S. Rushbrooke (North-Holland, Amsterdam, 1968), Chap. 5.
²⁴R. A. Sack, *Mol. Phys.* **2**, 8 (1959).
²⁵N. A. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, *J. Chem. Phys.* **21**, 1087 (1953).
²⁶M. J. Gillian, *Mol. Phys.* **49**, 461 (1983).